

DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

Enhancing tool-use dialogue flow via automatic preference pair construction

DiaTool-DPO: Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Formulates TA-LLM interactions as a 5-state MDP and 3 query-type trajectories

Automatically builds chosen/rejected dialogue pairs without human labels

Targets robust slot-filling and tool-rejection behavior for production deployment

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs must decide: ask follow-up, call tool, or reject tool call.

SFT with expert trajectories often under-trains dialogue control decisions.

Failure mode 1: misses follow-up questions when required slots are absent.

Failure mode 2: hallucinates tool invocations from incomplete user requests.

Failure mode 3: fails to reject out-of-scope requests (no suitable tool).

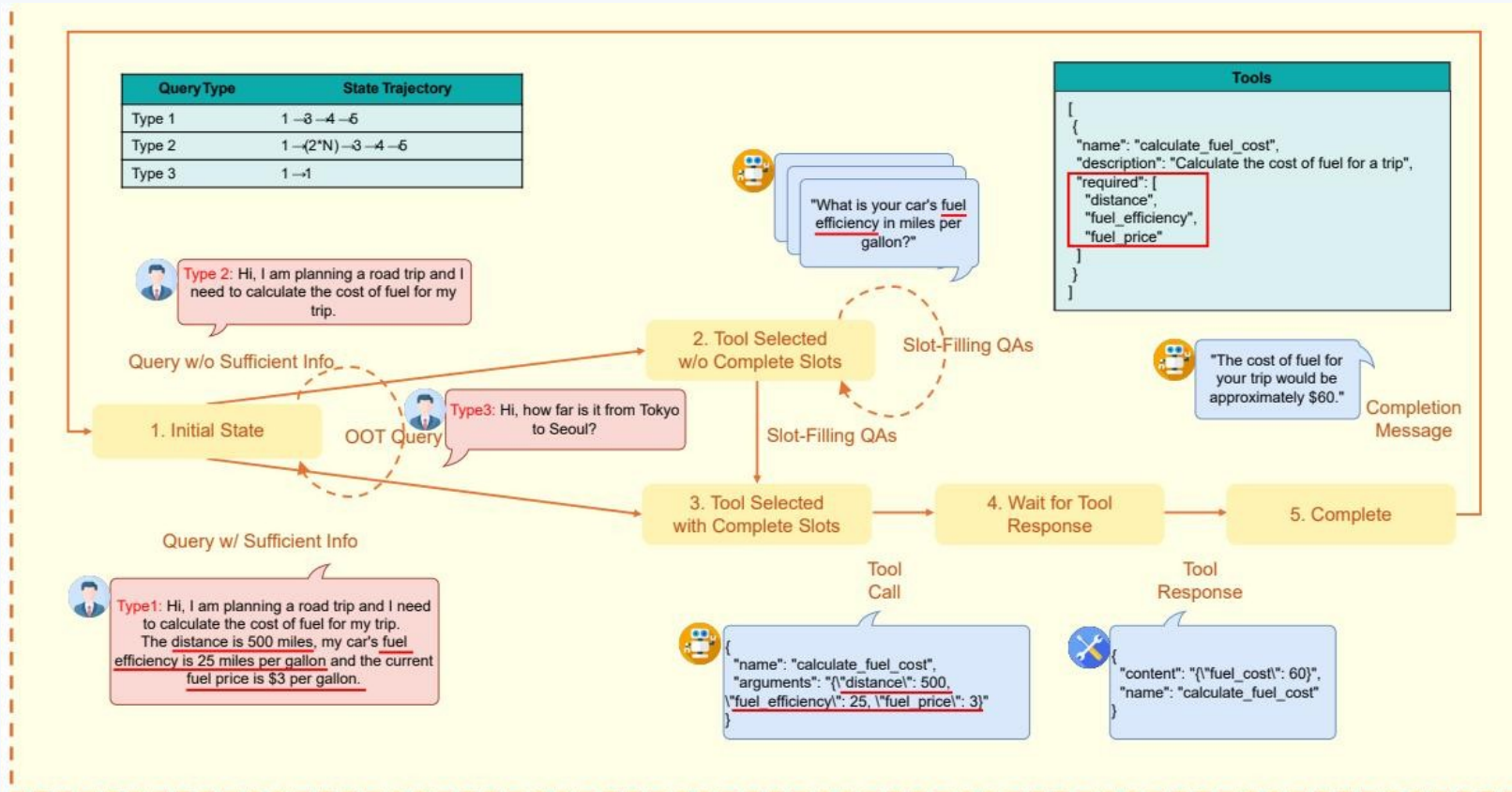
Why DiaTool-DPO?

Optimize preferences over multi-turn trajectories

Teach when to gather slots vs. call vs. reject

Improve production reliability of tool-augmented dialogue

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



State Definitions

1. Initial
 2. Tool selected, incomplete slots
 3. Tool selected, complete slots
 4. Wait for tool response
 5. Complete
- Out-of-scope requests remain in state 1 with rejection response.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Brief Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	User provides all required arguments; tool call proceeds directly.
Type 2	$1 \rightarrow (2 * N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	User omits required fields; model performs N rounds of slot-filling QA.
Type 3	$1 \rightarrow 1$	Requested functionality unavailable; model rejects tool call.

Trajectory taxonomy provides a deterministic scaffold for preference-pair generation.

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen traj.	Rejected traj.	Learning lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Observed Gains

Slot: 0.639 → 0.917 (+43.5%)

Relevance: 0.826 → 0.913 (+10.5%)

Largest gains occur in dialogue-control dimensions.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

- 5-state MDP + 3 query types create a formal control model for tool dialogue.
- Automatic chosen/rejected trajectory pairing removes human preference labeling overhead.
- SFT + DiaTool-DPO gives the best balanced performance (highest Macro Avg across models).
- DiaTool-DPO-only underperforms, showing DPO complements rather than replaces SFT.
- Method is language-agnostic and reaches near GPT-4o-level behavior on key dialogue tasks.